

Project Proposal: NutriGuide — AI Wellness & Nutrition Tracker

Domain: Health Assistance and Productivity

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Date: 2/4/2026

1. Problem Statement & Motivation

- Many college students and young adults want to eat healthier but struggle to stay consistent.
- Between classes, work, and social activities, meals are often rushed or unplanned.
- Over time, this makes it difficult to understand personal eating patterns.
- Most existing calorie-tracking apps are not ideal for long-term use.
 - Many require detailed manual entry.
 - Others focus heavily on weight loss instead of overall wellness.
 - Some feel overwhelmed after only a few days of use.
- As a result, users either stop tracking or stop paying attention to the data.
- From a practical standpoint, people want tools that fit into their routine without adding stress.
- Tracking food should feel useful, not like another assignment.
- Target users include:
 - College students

- Recent graduates
 - Young professionals interested in improving daily habits
- An agent-based AI system fits this problem well.
 - Nutrition tracking involves several connected steps.
 - These steps are easier to manage when different parts of the system handle specific tasks.
 - This leads to more reliable and consistent results.

2. Use Cases & Scenarios

Scenario 1: Daily Meal Logging

- A student types what they ate that day, such as “chicken wrap and iced coffee.”
- The system interprets the entry and separates it into individual food items.
- Nutrition information is retrieved and added together.
- A short summary shows calories and key nutrients for the day.

Scenario 2: Weekly Nutrition Review

- At the end of the week, the user asks for a summary.
- Past entries are reviewed to find patterns.
- Trends such as skipped meals or low protein intake are highlighted.
- Feedback focuses on realistic changes rather than strict rules.

Scenario 3: Wellness Monitoring

- Over time, eating habits become clearer.

- If long gaps between meals or very low intake appear, the system responds.
- Messages are supportive and focused on balance.
- No judgment or pressure is used.

3. Initial System Design

User Interface

- The project will use a simple web-based interface.
- The layout focuses on clarity and ease of use.
- Main elements include:
 - A meal entry box
 - Daily and weekly summaries
 - Basic charts
 - A feedback panel
 - Data management controls

Multi-Agent System Structure

- **Planner Agent**
 - Keeps track of workflow steps
 - Decides which agent acts next
- **Food Interpretation Agent**
 - Reads natural language food entries
 - Estimates portions when possible
- **Nutrition Database Agent**

- Searches public nutrition sources
 - Returns calorie and nutrient values
- **Calculation Agent**
 - Combines all nutrition data
 - Prepares summary statistics
- **Wellness Coaching Agent**
 - Translates numbers into practical advice
 - Emphasizes balance and flexibility
- **Safety Monitoring Agent**
 - Watches for extreme patterns
 - Displays health-related disclaimers
 - Suggests professional help when appropriate

Workflow

- Planner → Interpretation → Database → Calculation → Coaching → Safety → Results

Tools and Resources

- USDA and Nutritionix databases
- Local storage system
- Visualization libraries
- System logs for evaluation

4. Feasibility, Risks, & Privacy

Possible Challenges

- Food descriptions may be unclear.
- Users may forget to log meals.
- Some users may misunderstand feedback.

How These Issues Will Be Addressed

- The interface will allow quick editing and corrections.
- Reminders will be optional.
- Feedback will use simple, direct language.
- Safety checks will remain active at all times.

Privacy Considerations

- No personal identifying information is required.
- Only basic meal data is collected.
- All records remain on the user's device when possible.
- A full data deletion option is included.

Project Limitations

- Development time is limited.
- API request limits may apply.
- Testing will involve a small user group.

5. Project Timeline

Weeks 1–2

- Outline system structure
- Build early interface version
- Connect nutrition databases

Week 3 (Milestone I)

- Complete front-end features
- Test basic logging functions

Weeks 4–5 (Milestone II)

- Implement multi-agent workflow
- Add coaching and safety checks

Week 6 (Milestone III)

- Run pilot tests
- Gather user feedback

Week 7

- Improve usability
- Adjust feedback style

Week 8

- Finalize documentation
- Prepare presentation materials